

Global Constraints on Higgsing Localized Five-Dimensional Sectors in Compact M-Theory Vacua

Bernd Johannes Wuebben

New York, NY, USA

E-mail: wuebben@gmail.com

ABSTRACT: Complex deformations of noncompact canonical Calabi–Yau threefold singularities encode Higgs-branch renormalization-group flows of five-dimensional theories. We determine which such flows remain available when several singularities occur in one compact Calabi–Yau threefold. The local sectors are conifold points supporting free hypermultiplets and cones over del Pezzo surfaces of degrees seven and six engineering the rank-one E_2 and E_3 fixed points. We identify their deformation coordinates with flavor moment maps and derive a compact Abelian gauging matrix from wrapped-M2 charges and divisor pairings. On either E_3 branch (the minimal nilpotent orbit closure of $\mathfrak{sl}_3$ or that of $\mathfrak{sl}_2$), we prove that the invariant deformation-coordinate values attained on the triplet moment-map zero locus in the rigid description are precisely the kernel of this matrix. The geometric comparison applies to Δ -regular anticanonical hypersurfaces whose fan polytopes have unit edges and only unimodular triangles, unit squares, or one-interior-point pentagons and hexagons as two-faces, with unit dual edges at the singular faces. Assuming trivial canonical bundle, $h^1(\mathcal{O}) = 0$, and the $\partial\bar{\partial}$ -lemma on the crepant resolution, the same kernel is the image of the global first-order deformation space. This extends the charge description of conifold transitions to mixed transitions containing interacting and reducible Higgs sectors. A period argument supplies an all-orders necessary condition for analytic one-parameter smoothings on the nodal, E_2 , and $\mathfrak{sl}_2$ branches. We exhibit a mixed transition in which two conifold sectors and one E_2 sector can move only simultaneously along a common one-dimensional deformation, and a compact model with thirty locally smoothable sectors but no trajectory that Higgses all of them.

KEYWORDS: M-Theory, Supersymmetric Gauge Theory, Supergravity Models, Differential and Algebraic Geometry

Contents

1	Introduction and summary	1
2	Localized five-dimensional sectors and their Higgs coordinates	5
2.1	Geometric engineering and the meaning of the projection	6
2.2	The conifold sector	6
2.3	The E2 reduced component and nilpotent tangent	7
2.4	The two components of the E3 sector	8
2.5	Magnetic-quiver and Hasse-diagram check	9
3	Compact embedding and the five-dimensional gauging matrix	10
3.1	Vector fields and wrapped-M2 charges	10
3.2	Flavor cocharacters from del Pezzo roots	11
3.3	Branch profiles and the compact charge map	11
3.4	The projected moment-map locus	12
3.5	Compact supergravity qualification	13
4	Global deformation theorems	13
4.1	Links, Milnor fibers and branch lattices	13
4.2	A class of toric anticanonical hypersurfaces	14
4.3	The first-order image theorem	15
4.4	All-orders necessity	17
4.5	Integrability boundary	17
5	Compact examples: forbidden, mixed and obstructed deformations	18
5.1	A single E2 direction that disappears globally	18
5.2	A mixed smoothing requiring simultaneous local deformations	19
5.3	Thirty locally smoothable sectors with no common trajectory	21
5.4	Finite classification for polytopes with at most nine vertices	23
6	Normalizability and gravity-decoupling limits	23
6.1	Finite compact volume	23
6.2	Dynamical and background vectors in a local limit	24
6.3	Multistage decoupling	24
6.4	Complex deformations and the local L2 caveat	25
7	Discussion	25
7.1	Relation to generalized-symmetry gluing	26
7.2	Comparison with the compact del Pezzo–conifold precursor	26
7.3	Open problems	26

A	Local invariant rings and balanced representatives	27
A.1	E3 maximal-torus invariants	27
A.2	Balanced real moment maps	28
A.3	Abelian monopole sums for the E3 components	28
B	Integral charge data for the compact examples	29
B.1	The X9 divisor reduction	29
B.2	The X-circ toric and resolution input	29
B.3	Primitive vectors in X-circ	32
C	Reproducibility	32

1 Introduction and summary

Topology-changing transitions in Calabi–Yau compactifications admit a particularly direct interpretation in M-theory. At a conifold degeneration, M2-branes wrapped on exceptional curves become massless. Their charges are the homology classes of those curves, and condensation along the Higgs branch connects the resolved and deformed Calabi–Yau threefolds [1–4]. Contractions of compact del Pezzo surfaces lead instead to interacting five-dimensional fixed points [5–8]. These two kinds of transition are usually analyzed separately: the former in terms of charged hypermultiplets and the latter in terms of the Coulomb and Higgs branches of a local superconformal field theory.

This paper studies compact transitions in which both types of sector occur at once. We call a transition *cooperative* when its global deformation space contains a direction with nonzero components in several local sectors but contains no nonzero direction supported on any one of those sectors alone. The basic question is global. Suppose that a compact Calabi–Yau threefold contains isolated singularities p which, in their noncompact geometric-engineering limits, define five-dimensional theories $\mathcal{T}_p$. Each local theory can possess one or more Higgs-branch directions. Which combinations of these directions remain available when the singularities are embedded in the same compact threefold and coupled to the vector fields of the compactification?

There is no reason for the answer to factorize. A complex deformation of the compact threefold restricts to a deformation of every singular germ, but the image of the restriction map

$$\mathbb{T}_X^1 := \mathrm{Ext}_X^1(\Omega_X^1, \mathcal{O}_X) \longrightarrow \bigoplus_p T_{X,p}^1, \quad T_{X,p}^1 := \mathrm{Ext}_{\mathcal{O}_{X,p}}^1(\Omega_{X,p}^1, \mathcal{O}_{X,p}), \quad (1.1)$$

can be a proper subspace. The same phenomenon has a five-dimensional description. Reducing the M-theory three-form on harmonic two-forms gives Abelian vector fields A^I . Wrapped M2 states are electrically charged under these fields, while the compact embedding gauges

Cartan subgroups of the flavor symmetries of the interacting sectors. The corresponding moment-map equations correlate local Higgs operators that would be independent in isolated noncompact models.

We make this relation precise for compact Calabi–Yau threefolds whose isolated singularities are ordinary double points and anticanonical cones over del Pezzo surfaces of degrees seven and six. We write

$$S_7 = \text{Bl}_2 \mathbb{P}^2, \quad S_6 = \text{Bl}_3 \mathbb{P}^2. \quad (1.2)$$

The corresponding local sectors are a free hypermultiplet and the rank-one E_2 and E_3 fixed points, respectively. This notation avoids the common ambiguity between the degree of a del Pezzo surface and the number of blown-up points.

The local input is finer than a dimension count. In one complex structure, a nodal hypermultiplet with scalars $(h, \tilde{h})$ projects to the conifold deformation coordinate $t = h\tilde{h}$. For the E_2 theory, the invariant ring of its Higgs chiral ring under the complexified $SU(2)$ Cartan is

$$\mathbb{C}[\mathcal{H}_{E_2}]^{\mathbb{C}^*} \cong \frac{\mathbb{C}[J_0, U]}{(U^2, UJ_0)}. \quad (1.3)$$

This is the complete nonreduced miniversal base of the cone over S_7 . The reduced smoothing coordinate is the $SU(2)$ Cartan moment map J_0 ; the additional nilpotent tangent direction is the $U(1)$ current-multiplet operator U . For E_3 , taking invariants under the complexified maximal torus gives

$$\mathbb{C}[\mathcal{H}_{E_3}]^{T_{\mathbb{C}}} \cong \frac{\mathbb{C}[H_1, H_2, L]}{(H_1 L, H_2 L)}. \quad (1.4)$$

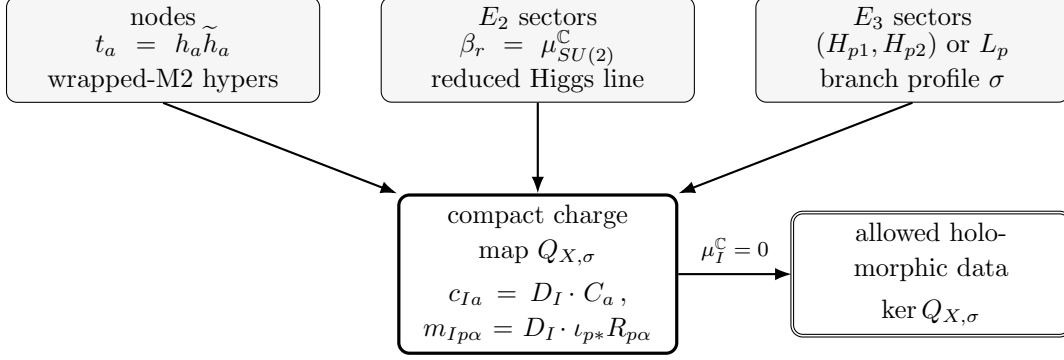
The plane (H_1, H_2) consists of the $SU(3)$ Cartan moment maps and the line L is the $SU(2)$ Cartan moment map. Equation (1.4) is the reducible miniversal base of the cone over S_6 . Its plane is the maximal-torus invariant image $\mathcal{B}_{\mathfrak{sl}_3} \cong \mathbb{C}^2$ of the minimal nilpotent orbit closure $\mathcal{H}_{\mathfrak{sl}_3} := \overline{\mathcal{O}_{\min}(\mathfrak{sl}_3)}$; its line is the corresponding image $\mathcal{B}_{\mathfrak{sl}_2} \cong \mathbb{C}$ of $\mathcal{H}_{\mathfrak{sl}_2} := \overline{\mathcal{O}_{\min}(\mathfrak{sl}_2)}$. This description keeps the two full E_3 Higgs components and their invariant images separate rather than combining them into a single tangent-space count. The local rings follow from the known Higgs chiral rings [9]; magnetic quivers provide an independent component and Hilbert-series check [10–12].

Charge constraints and the local-to-global deformation map

Let $\hat{X} \rightarrow X$ be a crepant resolution and let D_I be compact divisor classes. At a node, denote the exceptional curve by C_a . At an E_2 or E_3 point, let $R_{p\alpha}$ be a root in the summand of $K_{S_p}^\perp$ selected by the chosen Higgs component. Here $K_{S_p}^\perp$ is the orthogonal complement of the canonical class in $H_2(S_p; \mathbb{Z})$ for the intersection pairing. The compact matrix is

$$c_{Ia} = D_I \cdot C_a, \quad m_{Ip\alpha} = D_I \cdot \iota_{p*} R_{p\alpha}. \quad (1.5)$$

The first set of entries are the electric charges of wrapped M2-branes. The second set are the cocharacters specifying how A^I gauges the local flavor Cartans. Thus the complex moment



The same divisor–curve pairing is the wrapped-M2 electric charge, the flavor-cocharacter gauging, and the dual of the geometric local-to-global map.

Figure 1. The compact Higgs-gluings mechanism on a fixed E_3 branch profile. The displayed kernel is the image in invariant complex moment-map coordinates. Balanced representatives lift it to the zero level of the full triplet moment maps; integration to a finite complex deformation is a separate question.

maps take the form

$$\mu_I^C = \sum_a c_{Ia} t_a + \sum_r m_{Ir}^{(7)} \beta_r + \sum_p (m_{Ip1} H_{p1} + m_{Ip2} H_{p2} + m_{Ip0} L_p), \quad (1.6)$$

with $H_{p1} L_p = H_{p2} L_p = 0$ at each E_3 point.

A *branch profile* σ chooses the full component $\mathcal{H}_{\mathfrak{sl}_2}$ or $\mathcal{H}_{\mathfrak{sl}_3}$ at each E_3 point; $\mathcal{B}_\sigma$ is the product of their invariant images with the nodal and reduced E_2 lines. Let $\mathcal{M}_{H,\sigma}^{\text{rigid}}$ denote the product of the selected local Higgs cones, restricted to the triplet-moment-map zero locus for the effective compact Abelian group and then divided by that group. Restricting (1.6) to the invariant coordinates gives a linear complex map $Q_{X,\sigma}$. For the geometric equality below, X is a Δ -regular anticanonical hypersurface associated with an admissible four-dimensional reflexive polytope as defined in Section 4.2, and it satisfies (4.8). Our principal result is

$$\boxed{\text{Im}\left(\mathcal{M}_{H,\sigma}^{\text{rigid}} \longrightarrow \mathcal{B}_\sigma\right) = \ker(Q_{X,\sigma}) = \text{Im}\left(\mathbb{T}_X^1 \longrightarrow \bigoplus_p T_{X,p}^1\right) \cap \mathcal{B}_\sigma.} \quad (1.7)$$

The first equality is a five-dimensional moment-map statement. The second is the geometric local-to-global deformation theorem. Equation (1.7) therefore identifies the geometric kernel with a physical charge and gauging constraint, rather than attaching a field-theory interpretation to a dimension computed after the fact.

Figure 1 summarizes the three local inputs and the common compact charge map.

The word “image” in (1.7) is essential. A local deformation base records invariant Higgs-branch coordinates, not the full hyperkähler cone or its metric. Likewise, the right-hand

side is a complex linear space on a fixed reduced profile. We do not identify it with the full quaternionic-Kähler scalar manifold of the compact supergravity theory.

The real moment maps do not impose an additional inequality on this projected image. Every value of t_a , β_r , (H_{p1}, H_{p2}) , or L_p has a balanced representative for which the corresponding local real Cartan moment maps vanish. Consequently every point of the complex kernel lifts to the zero-level triplet-moment-map locus. This also explains why signs in a homology relation do not obstruct Higgsing: phases reside in the elementary complex scalars, while their absolute values can be balanced.

The geometric theorem has two levels. Under the stated compactness, Hodge and admissibility hypotheses, (1.7) is an exact first-order statement. A separate vanishing-period argument constrains genuine analytic arcs to all orders, including arcs tangent to a discriminant to arbitrarily high order. It implies a full-support relation for nodes, E_2 points and E_3 points smoothed through $\mathcal{H}_{\mathfrak{sl}_2}$. The two-dimensional component $\mathcal{H}_{\mathfrak{sl}_3}$ requires a two-period extension that is not claimed here. Conversely, the general integration of an allowed first-order class remains open; it is proved in the explicit positive example below.

Three compact phenomena

The theorem produces three qualitatively different effects.

First, a local Higgs direction can disappear completely after compact embedding. We give a compact model X_7 with one E_2 cone point. The reduced local theory has its $SU(2)$ Higgs component, but the compact kernel is zero. Hence no global first-order deformation activates it and no analytic smoothing exists.

Second, local sectors can Higgs only cooperatively. Our main explicit model X_9 contains two nodes and one E_2 point. Neither the two nodal sectors alone nor the E_2 sector alone has a nonzero global direction. In a primitive compact vector basis the effective charge matrix is

$$Q_{X_9}^{\text{eff}} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad \ker Q_{X_9}^{\text{eff}} = \mathbb{C}(-1, -1, 1). \quad (1.8)$$

The unique projected Higgs direction moves all three sectors together. An explicit family integrates it. The Hodge numbers change as

$$(h^{1,1}, h^{2,1}) = (6, 122) \longrightarrow (3, 123). \quad (1.9)$$

Two vectors are Higgsed by the rank-two gauging, while the third disappearing vector is the neutral rank-one E_2 Coulomb direction represented by the exceptional divisor. The surviving deformation supplies one neutral hypermultiplet, in agreement with (1.9).

Third, local smoothability need not imply simultaneous global Higgsability. A compact model X° contains 26 nodes, two E_2 points and two E_3 points. Every germ is locally smoothable. For all four ordered choices of $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ components at the two E_3 points, however, explicit primitive compact vectors force two nodal and both E_2 deformation coordinates to vanish. The projected kernels have dimensions 9, 10, 10, 12, so many partial Higgs directions

survive, but no profile contains a trajectory that activates all thirty sectors. The vanishing-period theorem promotes this conclusion beyond tangent space on every profile, because any smoothing must move the two forced nodes and both E_2 points whatever components it selects at the E_3 points. Here the superscript circle is part of the model name and does not denote the polar operation introduced in (4.7).

Relation to previous work and scope

Compact conifold charge matrices and their five-dimensional gauged-supergravity actions were developed in [4]. Malyshev’s compact quintic supplied the closest geometric precursor: a del Pezzo root curve is homologous to four conifold curves, reducing the number of del Pezzo deformations [13]. Local E_n Higgs branches and magnetic quivers are treated in [9–12, 14]. De Marco, Del Zotto, Grimminger and Sangiovanni develop the local Higgs-branch stratification of five-dimensional fixed points from canonical singularities [15]. The present problem is the subsequent compact one: which invariant directions survive when several such sectors share the dynamical vectors of a single compact threefold. Compact gluing and multistage decoupling are analyzed in [16, 17].

Our result extends the mixed geometry in [13]. It gives a uniform theorem for free and interacting localized sectors and derives its matrix as the electric-charge and flavor-cocharacter gauging map. It also retains the two components of the E_3 Higgs cone, identifies the full scheme-level E_2/E_3 projections, and separates the exact first-order image theorem from the all-orders obstruction. The cooperative and obstructed compact examples test each part of this structure. None of these field-theoretic or branchwise statements is contained in the one-example deformation count of [13].

All compact harmonic modes used in the construction have finite norm at finite volume, so the full matrix in (1.6) is dynamical in the compact theory. In a limit that decouples gravity, only vector combinations with finite nonzero kinetic norm remain dynamical. The matrix is then projected to $Q_{\text{dyn}} = P_{\text{dyn}} Q_X$. Gravity may therefore decouple before all local sectors factorize; conversely, a fully factorized local limit removes the shared rows. We do not claim that the compact constraint survives every decoupling limit, and no example-specific Kähler scaling ray is needed for the finite-volume theorem.

The paper is organized as follows. Section 2 establishes the local operator dictionaries and magnetic-quiver checks. Section 3 derives the compact charge and gauging map. Section 4 states the geometric first-order and all-orders theorems. Section 5 presents the absent, cooperative and obstructed Higgsing examples. Section 6 separates the compact result from its possible rigid limits, and Section 7 discusses extensions and open problems.

2 Localized five-dimensional sectors and their Higgs coordinates

The compact argument requires the holomorphic flavor-moment-map coordinates that coincide with geometric deformation parameters. We identify these coordinates as affine schemes;

singularity	local 5d sector	$\mathcal{B}_{\text{red}}$	invariant moment-map coordinates
ordinary double point	free hypermultiplet	$\mathbb{C}_t$	$t = h\tilde{h}$
cone over $S_7 = \text{Bl}_2 \mathbb{P}^2$	rank-one E_2 theory	$\mathbb{C}_\beta$	$\beta = J_0$, the $SU(2)$ Cartan moment map
cone over $S_6 = \text{Bl}_3 \mathbb{P}^2$	rank-one E_3 theory	$\mathbb{C}_{H_1, H_2}^2 \cup_0 \mathbb{C}_L$	(H_1, H_2) for $SU(3)$ and L for $SU(2)$

Table 1. The local holomorphic projection used in the compact theorem. The E_2 miniversal base also contains an additional nilpotent tangent direction generated by U ; it is not an ordinary reduced Higgs trajectory.

the remaining Higgs-branch data of the rank-one theories will not enter the global charge map.

2.1 Geometric engineering and the meaning of the projection

M-theory on a noncompact canonical Calabi–Yau threefold singularity Y defines a five-dimensional theory $\mathcal{T}_Y$ [5, 7, 8]. A crepant resolution of Y parametrizes its Coulomb data, while normalizable complex deformations encode Higgs-branch data. This dictionary has been developed in many examples and can be refined to the stratification by symplectic leaves [12, 15].

Two qualifications will be important. First, the base $\text{Def}(Y)$ of complex deformations is not itself the hyperkähler Higgs branch. In the examples below it is obtained from a chosen complex structure on the Higgs cone by taking affine invariants under a complexified flavor torus. It therefore records Cartan moment-map values but forgets the remaining orbit coordinates. Second, strict L^2 normalizability in an isolated noncompact metric is a separate analytic question. The compact theorem uses limits of modes on a compact threefold, where their norms are finite. We return to noncompact limits in Section 6.

The three local dictionaries are summarized in Table 1. Here $\mathcal{B}_{\text{red}}$ denotes the reduced geometric deformation component selected by an ordinary complex Higgs vev.

2.2 The conifold sector

For the ordinary double point

$$Y_0 = \{xy - zw = 0\} \subset \mathbb{C}^4, \quad (2.1)$$

the small resolution replaces the singular point by a curve $C \cong \mathbb{P}^1$. An M2-brane wrapping C gives a five-dimensional hypermultiplet. In one complex structure let $(h, \tilde{h})$ be its two complex scalars, with opposite charges under the relevant $U(1)$. The local complex moment map and the conifold deformation coordinate are the same invariant,

$$t = h\tilde{h}, \quad Y_t = \{xy - zw = t\}. \quad (2.2)$$

This local map can also be obtained from the type-IIA description of the rank-zero theory [18].

For later use, every $t \in \mathbb{C}$ has a representative with zero real moment map. If $t = |t|e^{i\vartheta}$, take

$$h = \sqrt{|t|}e^{i\vartheta}, \quad \tilde{h} = \sqrt{|t|}. \quad (2.3)$$

Then $h\tilde{h} = t$ and $|h|^2 - |\tilde{h}|^2 = 0$. This elementary observation is the local reason that arbitrary complex coefficients in a homology relation can solve the full triplet of moment-map equations.

2.3 The E_2 reduced component and nilpotent tangent

The cone over S_7 engineers the rank-one E_2 fixed point with flavor algebra

$$\mathfrak{e}_2 = \mathfrak{su}(2) \oplus \mathfrak{u}(1). \quad (2.4)$$

In a chosen complex structure, the Higgs chiral ring is generated by an $SU(2)$ moment-map triplet (J_+, J_0, J_-) and a further current-multiplet operator U , denoted P in [9]:

$$\mathbb{C}[\mathcal{H}_{E_2}] = \frac{\mathbb{C}[J_+, J_0, J_-, U]}{(J_0^2 - J_+J_-, U^2, UJ_+, UJ_0, UJ_-)}. \quad (2.5)$$

The reduction is the A_1 surface

$$\mathcal{H}_{E_2, \text{red}} = \{J_0^2 = J_+J_-\} \cong \mathbb{C}^2/\mathbb{Z}_2, \quad (2.6)$$

and U defines an embedded point at its vertex.

The complexified $SU(2)$ Cartan has weights $(2, 0, -2, 0)$ on (J_+, J_0, J_-, U) . A weight-zero monomial in the reduced ring is a power of J_0 , because

$$J_+^a J_0^b J_-^a = (J_+J_-)^a J_0^b = J_0^{2a+b}. \quad (2.7)$$

Every nonzero monomial containing U is proportional to U . It follows that

$$\mathbb{C}[\mathcal{H}_{E_2}]^{\mathbb{C}^*} = \frac{\mathbb{C}[J_0, U]}{(U^2, UJ_0)}. \quad (2.8)$$

Altmann's miniversal base for the degree-seven cone is

$$\text{Def}(C(S_7)) = \text{Spec} \frac{\mathbb{C}[s_2, s_1]}{(s_1^2, s_1 s_2)} \quad (2.9)$$

[19]. The root marking from the crepant resolution fixes the dictionary, up to nonzero normalization,

$$\beta = s_2 \longleftrightarrow J_0, \quad \alpha = s_1 \longleftrightarrow U. \quad (2.10)$$

Thus the complete nonreduced schemes agree, not only their tangent dimensions. The reduced line $\alpha = 0$ is the physical finite Higgs component. Since $U^2 = UJ_0 = 0$, the U direction exists

over dual numbers but vanishes at every ordinary complex point; it cannot be counted as a second finite branch.

The Cartan interpretation is explicit in the double-cover coordinates:

$$J_+ = a^2, \quad J_0 = ab, \quad J_- = b^2, \quad (a, b) \sim (-a, -b). \quad (2.11)$$

The Cartan acts as $(a, b) \mapsto (\lambda a, \lambda^{-1} b)$ and has moment maps

$$\mu_{E_2}^{\mathbb{C}} = ab = \beta, \quad \mu_{E_2}^{\mathbb{R}} = \frac{1}{2}(|a|^2 - |b|^2). \quad (2.12)$$

The choice $a = \sqrt{|\beta|} e^{i \arg \beta}$, $b = \sqrt{|\beta|}$ realizes any complex β while setting the real moment map to zero.

2.4 The two components of the E_3 sector

The cone over S_6 engineers the rank-one E_3 fixed point with flavor algebra

$$\mathfrak{e}_3 = \mathfrak{su}(3) \oplus \mathfrak{su}(2). \quad (2.13)$$

Its reduced Higgs branch is the union

$$\mathcal{H}_{E_3} = \underbrace{\overline{\mathcal{O}_{\min}(\mathfrak{sl}_3)}}_{\mathcal{H}_{\mathfrak{sl}_3}} \cup_{\{0\}} \underbrace{\overline{\mathcal{O}_{\min}(\mathfrak{sl}_2)}}_{\mathcal{H}_{\mathfrak{sl}_2}}. \quad (2.14)$$

The two components have quaternionic dimensions two and one. In terms of the complex flavor moment maps $\Phi \in \mathfrak{sl}_3$ and $\tilde{T} \in \mathfrak{sl}_2$, the chiral-ring relations are

$$\Phi^2 = 0, \quad \text{tr}(\tilde{T}^2) = 0, \quad \Phi^i_j \tilde{T}^{\dot{\alpha}\dot{\beta}} = 0 \quad (2.15)$$

[9]. The mixed relation says scheme-theoretically that positive-degree functions on different components annihilate one another. The intersection is one reduced point.

Let (H_1, H_2) be root-basis Cartan coordinates of Φ and let L be the Cartan coordinate of $\tilde{T}$. The maximal-torus invariant rings of the two minimal nilpotent orbit closures are respectively $\mathbb{C}[H_1, H_2]$ and $\mathbb{C}[L]$. Taking invariants preserves the fiber product, because a complex torus is linearly reductive. Hence

$$\mathbb{C}[\mathcal{H}_{E_3}]^{T_{\mathbb{C}}} = \mathbb{C}[H_1, H_2] \times_{\mathbb{C}} \mathbb{C}[L] \cong \frac{\mathbb{C}[H_1, H_2, L]}{(H_1 L, H_2 L)}. \quad (2.16)$$

We reserve $\mathcal{H}_{\mathfrak{sl}_3}$ and $\mathcal{H}_{\mathfrak{sl}_2}$ for the full orbit closures and denote their invariant-coordinate images by

$$\mathcal{B}_{\mathfrak{sl}_3} := \text{Spec } \mathbb{C}[H_1, H_2] \cong \mathbb{C}^2, \quad \mathcal{B}_{\mathfrak{sl}_2} := \text{Spec } \mathbb{C}[L] \cong \mathbb{C}. \quad (2.17)$$

The miniversal base of the degree-six cone is

$$\text{Def}(C(S_6)) = \text{Spec } \frac{\mathbb{C}[s_1, s_2, s_3]}{(s_1 s_3, s_2 s_3)}. \quad (2.18)$$

The compatible root marking gives

$$(s_1, s_2) \longleftrightarrow (H_1, H_2) \quad \text{on } \mathcal{B}_{\mathfrak{sl}_3}, \quad s_3 \longleftrightarrow L \quad \text{on } \mathcal{B}_{\mathfrak{sl}_2}. \quad (2.19)$$

The mixed equations in (2.16) matter in the compact problem: at one E_3 point, an ordinary Higgs trajectory chooses the $SU(3)$ or $SU(2)$ moment maps, but not both.

With $K_{S_6}^\perp$ denoting the orthogonal complement of the canonical class in $H_2(S_6; \mathbb{Z})$ for the intersection pairing, the corresponding root decomposition inside the del Pezzo exceptional divisor is

$$K_{S_6}^\perp \cong A_2 \oplus A_1, \quad R_1 = e_1 - e_2, \quad R_2 = e_2 - e_3, \quad R_0 = \ell - e_1 - e_2 - e_3. \quad (2.20)$$

Here (R_1, R_2) span A_2 and R_0 spans its orthogonal A_1 . Root-basis coordinates are used throughout so that a charge functional with values (m_1, m_2) on the simple roots couples as $m_1 H_1 + m_2 H_2$, without an unwritten inverse Cartan matrix.

As in the E_2 case, arbitrary complex Cartan values admit representatives with zero real moment maps. One may use the rank-one Atiyah–Drinfeld–Hitchin–Manin (ADHM) presentation of a minimal nilpotent orbit closure: write $z_i = q_i \tilde{q}_i$, impose $\sum_i z_i = 0$, and choose $|q_i| = |\tilde{q}_i| = \sqrt{|z_i|}$ with phases reproducing z_i . This sets the ADHM and flavor real moment maps to zero pairwise. The $\mathfrak{sl}_2$ component is the same construction with two pairs.

2.5 Magnetic-quiver and Hasse-diagram check

The $\mathfrak{sl}_3$ and $\mathfrak{sl}_2$ components have the affine- A_2 triangle and affine- A_1 double-edge magnetic quivers, after removing the decoupled diagonal $U(1)$ [10–12]; the two-cone structure of the E_3 Higgs branch goes back to [5, 6]. Their Coulomb-branch dimensions are $3 - 1 = 2$ and $2 - 1 = 1$, agreeing with (2.19). With $x = t^2$, the Abelian monopole formula [20] gives

$$H_{\mathfrak{sl}_2}(x) = \frac{1+x}{(1-x)^2} = \sum_{n \geq 0} (2n+1)x^n, \quad (2.21)$$

$$H_{\mathfrak{sl}_3}(x) = \frac{1+4x+x^2}{(1-x)^4} = \sum_{n \geq 0} (n+1)^3 x^n. \quad (2.22)$$

Since the components meet in a reduced point,

$$H_{E_3}(x) = H_{\mathfrak{sl}_3}(x) + H_{\mathfrak{sl}_2}(x) - 1. \quad (2.23)$$

The coefficient of x is $8+3=11$, the dimension of the $SU(3) \times SU(2)$ moment-map generators. This is independent of the compact deformation calculation.

The component data relevant below are collected in Table 2. Each minimal nilpotent orbit closure has only its open orbit and the origin as symplectic leaves. A generic point therefore has no transverse interacting endpoint; this local statement does not remove bulk vectors or other compact sectors.

component	magnetic quiver	$\dim_{\mathbb{H}}$	coordinate	generic continuous stabilizer
$\mathcal{H}_{\mathfrak{sl}_3}$	affine A_2	2	(H_1, H_2)	$U(1) \times SU(2)$
$\mathcal{H}_{\mathfrak{sl}_2}$	affine A_1	1	L	$SU(3)$

Table 2. Independent local data for the two E_3 Higgs components. The $SU(2)$ (respectively $SU(3)$) factor complementary to the active $\mathfrak{sl}_3$ (respectively $\mathfrak{sl}_2$) orbit is unbroken. The table lists the full continuous generic stabilizer and omits only finite subgroups.

3 Compact embedding and the five-dimensional gauging matrix

The physical matrix correlating the coordinates of Section 2 has two types of columns. Exceptional curves at nodes give electrically charged wrapped-M2 hypermultiplets. Roots in an exceptional del Pezzo surface determine how compact Abelian vectors embed into the flavor Cartan of the interacting fixed point. Both columns are obtained by evaluating curve classes on compact divisors.

3.1 Vector fields and wrapped-M2 charges

Let X be a compact singular Calabi–Yau threefold and let

$$\pi : \widehat{X} \longrightarrow X \quad (3.1)$$

be a projective crepant resolution. Choose harmonic two-forms ω_I on $\widehat{X}$ whose cohomology classes span the vector directions relevant to the transition. The M-theory three-form has the expansion

$$C_3 = \sum_I A^I \wedge \omega_I + \dots \quad (3.2)$$

If an M2-brane wraps a curve $C \subset \widehat{X}$, its worldline coupling is

$$\int_{\mathbb{R} \times C} C_3 = \sum_I \left(\int_C \omega_I \right) \int_{\mathbb{R}} A^I. \quad (3.3)$$

For a divisor class D_I dual to ω_I , the electric charge is therefore

$$c_I(C) = \int_C \omega_I = D_I \cdot C. \quad (3.4)$$

At a resolved node a , let $C_a \cong \mathbb{P}^1$ denote the exceptional curve. The wrapped M2-brane is the massless hypermultiplet of the conifold transition, with charge vector

$$c_{Ia} = D_I \cdot C_a. \quad (3.5)$$

Thus the map

$$\mathbb{Z}^{N_{\text{node}}} \longrightarrow H_2(\widehat{X}; \mathbb{Z})/\text{tors}, \quad e_a \longmapsto [C_a] \quad (3.6)$$

is simultaneously the homology map of the resolution and the electric charge map of the transition states. This is the standard compact conifold description [2, 4].

3.2 Flavor cocharacters from del Pezzo roots

Suppose the exceptional locus over p contains $S_{7,p}$ or $S_{6,p}$. The local non-Abelian flavor lattice is the root lattice inside $K_{S_p}^\perp$, the orthogonal complement of K_{S_p} in $H_2(S_p; \mathbb{Z})$ for the intersection form. Let

$$\iota_p : S_p \hookrightarrow \widehat{X} \quad (3.7)$$

be the inclusion and let $R_{p\alpha}$ be a root. The restriction $D_I|_{S_p}$ assigns degrees to the M2 states supported on curves in S_p . For E_2 , with $R = e_1 - e_2$, the difference of the charges of the two exceptional-curve states is

$$c_I(e_1) - c_I(e_2) = D_I \cdot \iota_{p*}(e_1 - e_2). \quad (3.8)$$

This is the differential of the homomorphism from the compact $U(1)$ generated by A^I into the $SU(2)$ flavor torus. More generally, define

$$m_{Ip\alpha} = \langle D_I|_{S_p}, R_{p\alpha}^\vee \rangle \sim D_I \cdot \iota_{p*} R_{p\alpha}. \quad (3.9)$$

The proportionality allows for the conventional common normalization between a primitive geometric root and a Cartan generator. All algebras here are simply laced. We fix compatible root-marked moment-map coordinates throughout. Changing a root/coroot normalization or orientation changes the corresponding coordinate and matrix column together. The resulting kernels are identified by this invertible coordinate change; their coordinate vectors are not literally unchanged.

For E_2 there is one reduced root coordinate β_p . For E_3 , the A_2 component uses (R_{p1}, R_{p2}) and the A_1 component uses R_{p0} , as in (2.20). Gauging a flavor isometry couples the vector-multiplet auxiliary triplet to the corresponding current-multiplet moment map [21]. Since moment maps add under diagonal gauging, the compact complex moment maps are

$$\mu_I^\mathbb{C} = \sum_a c_{Ia} t_a + \sum_r m_{Ir}^{(7)} \beta_r + \sum_p (m_{Ip1} H_{p1} + m_{Ip2} H_{p2} + m_{Ip0} L_p). \quad (3.10)$$

This conclusion uses only the flavor current multiplets of the interacting SCFT; it does not require a weakly coupled Lagrangian description of the E_2 or E_3 fixed point.

3.3 Branch profiles and the compact charge map

At each E_3 point choose either the full Higgs component $\mathcal{H}_{\mathfrak{sl}_2}$ or $\mathcal{H}_{\mathfrak{sl}_3}$. A collection of choices is a *branch profile* σ . The associated local coordinate space uses the invariant images of (2.17):

$$\mathcal{B}_\sigma = \bigoplus_a \mathbb{C}_{t_a} \oplus \bigoplus_r \mathbb{C}_{\beta_r} \oplus \bigoplus_p \mathcal{B}_{p,\sigma_p}, \quad \mathcal{B}_{p,\mathfrak{sl}_2} = (\mathcal{B}_{\mathfrak{sl}_2})_p = \mathbb{C}_{L_p}, \quad \mathcal{B}_{p,\mathfrak{sl}_3} = (\mathcal{B}_{\mathfrak{sl}_3})_p = \mathbb{C}_{H_{p1}, H_{p2}}^2. \quad (3.11)$$

Let Λ_{vec} be the lattice generated by the compact Abelian vector directions A^I , equivalently by their divisor classes D_I modulo the integral principal-divisor relations; the D_I are taken to span $H^2(\widehat{X}; \mathbb{Q})$, which on the admissible class of Section 4.2 the restricted ambient toric

divisors do. Its dual records one complex moment-map equation for each vector direction. The restriction of (3.10) to a profile is therefore the linear map

$$Q_{X,\sigma} : \mathcal{B}_\sigma \longrightarrow \Lambda_{\text{vec}}^\vee \otimes \mathbb{C}. \quad (3.12)$$

The same map can be defined before complexification. Let Λ_{p,σ_p}^H be the root sublattice selected by the local component, and let a nodal summand be generated by $[C_a]$. Then

$$\begin{aligned} \Lambda_\sigma^H &= \bigoplus_a \mathbb{Z}[C_a] \oplus \bigoplus_p \Lambda_{p,\sigma_p}^H, \\ Q_{X,\sigma}^{\text{geo}} : \Lambda_\sigma^H &\longrightarrow H_2(\widehat{X}; \mathbb{Z})/\text{tors}, \quad \gamma \longmapsto \sum_p \iota_{p*} \gamma_p. \end{aligned} \quad (3.13)$$

Evaluating (3.13) on the divisor classes gives the matrix in (3.12). In other words, the geometric local-to-global map and the physical embedding of the local electric and flavor charges are dual descriptions of one lattice homomorphism.

In the toric hypersurface examples, the finite matrix is initially computed using ambient divisor classes. The geometric comparison reviewed in Section 4 proves that, under the admissibility hypotheses used here, its branch-restricted kernel equals the complete topological kernel. Thus the finite presentation does not miss a continuous constraint. Questions about a possible finite index between the printed toric lattice and the full integral cohomology lattice affect global forms and line operators, but not the rational kernel or the continuous Higgs selection rule.

3.4 The projected moment-map locus

Let G_{eff} be the compact Abelian group acting effectively on the transition sectors through (3.10). In the rigid zero-level description, form the triplet-moment-map quotient of the product of the local Higgs cones on a fixed profile. We use $\mathcal{M}_{H,\sigma}^{\text{rigid}}$ for this quotient and retain only its map to the invariant coordinates (3.11).

Proposition 3.1 (physical kernel). *For every branch profile σ , the image of the rigid zero-level Higgs locus in its local deformation coordinates is*

$$\text{Im}(\mathcal{M}_{H,\sigma}^{\text{rigid}} \longrightarrow \mathcal{B}_\sigma) = \ker(Q_{X,\sigma}). \quad (3.14)$$

Proof. If a point solves the complex moment-map equations, equation (3.10) states exactly that its invariant coordinate vector $b \in \mathcal{B}_\sigma$ obeys $Q_{X,\sigma}b = 0$. This proves one inclusion.

Conversely, take $b \in \ker Q_{X,\sigma}$. Lift every nodal coordinate t_a by (2.3). Lift every E_2 coordinate β_r by (2.11) with equal absolute values. On an E_3 component, use the balanced rank-one ADHM representatives described after (2.20). Every local real Cartan moment map then vanishes separately. The total real moment maps consequently vanish, while the total complex moment maps vanish because $Q_{X,\sigma}b = 0$. This constructs a point of the zero-level triplet-moment-map locus above every b in the complex kernel. Since the coordinates in $\mathcal{B}_\sigma$ are gauge invariant, taking the quotient does not change the image. $\square$

If the selected local components have total quaternionic dimension d_{loc} and a rank- s Abelian group acts locally freely, the familiar count gives

$$\dim_{\mathbb{H}} \mathcal{M}_H = d_{\text{loc}} - s = \dim_{\mathbb{C}} \ker Q_{X,\sigma}. \quad (3.15)$$

Equation (3.14) is stronger than this count: it specifies which combinations of coordinates survive and remains meaningful at profiles where the action has nontrivial stabilizers. It also avoids the incorrect identification of the full Higgs branch with a complex vector space.

3.5 Compact supergravity qualification

The microscopic charge assignment (3.4) and flavor embedding (3.9) are exact compactification data. Proposition 3.1 is stated in the rigid hyperkähler-cone description at zero moment-map level. The scalar manifold of five-dimensional $\mathcal{N} = 1$ supergravity is instead quaternionic Kähler, and its detailed metric receives gravitational and strong-coupling corrections. We use the rigid statement only to identify the holomorphic Higgs-deformation projection and its compact gauging constraint. We do not claim that it constructs the complete finite-Planck-mass scalar manifold.

This distinction does not turn the matrix into a background flavor constraint. At finite compact volume the fields A^I in (3.2) have finite kinetic terms and are dynamical. Their auxiliary-field equations impose the compact moment maps. What changes in a noncompact limit is the set of rows that remain dynamical, as discussed in Section 6.

4 Global deformation theorems

The physical matrix of Section 3 has an independent geometric characterization. We formulate it in terms of the topology of the singularity links and Milnor fibers. This section also distinguishes the exact first-order statement from the all-orders condition on an analytic smoothing arc.

4.1 Links, Milnor fibers and branch lattices

Let Σ_p be the link of the isolated singularity (X, p) . For a smoothing component σ_p , let F_{p,σ_p} be a Milnor fiber and define the vanishing subspace

$$V_{p,\sigma_p} = \ker [H_2(\Sigma_p; \mathbb{Q}) \longrightarrow H_2(F_{p,\sigma_p}; \mathbb{Q})]. \quad (4.1)$$

The spaces used in this paper are

(X, p)	$H_2(\Sigma_p; \mathbb{Q})$	V_{p,σ_p}
node	$\mathbb{Q}$	$\mathbb{Q}$
$C(S_7)$	$K_{S_7}^\perp \otimes \mathbb{Q}$	$A_1 \otimes \mathbb{Q}$
$C(S_6), \mathcal{H}_{\mathfrak{sl}_2}$	$K_{S_6}^\perp \otimes \mathbb{Q}$	$A_1 \otimes \mathbb{Q}$
$C(S_6), \mathcal{H}_{\mathfrak{sl}_3}$	$K_{S_6}^\perp \otimes \mathbb{Q}$	$A_2 \otimes \mathbb{Q}$

(4.2)

For a del Pezzo cone, the link is the unit-circle bundle of K_{S_p} and the Gysin sequence gives

$$H_2(\Sigma_p; \mathbb{Q}) \cong K_{S_p}^\perp \subset H_2(S_p; \mathbb{Q}), \quad (4.3)$$

where the superscript $\perp$ refers to the intersection pairing. The root subspaces in (4.2) are the cycles that die in the corresponding Milnor fiber (Proposition 2.1 and Section 2.4 of [22], where the three sweeping Fano threefolds are identified).

Choose a branch profile σ and set

$$K_\sigma(X) = \ker \left[\bigoplus_p H_2(\Sigma_p; \mathbb{Q}) \longrightarrow H_2(\widehat{X}; \mathbb{Q}) \right] \cap \bigoplus_p V_{p, \sigma_p}. \quad (4.4)$$

The map sends a nodal generator to its exceptional curve class and a del Pezzo link class to its pushforward from the del Pezzo exceptional divisor. Thus (4.4) is the rational kernel of (3.13).

Lemma 4.1 (local deformation–link comparison). *Let $\widehat{X}_p \rightarrow (X, p)$ be the chosen crepant local resolution, with exceptional set E_p , and put*

$$K_p^{\text{cr}} = \ker \left[H_{E_p}^2(\widehat{X}_p; \Omega_{\widehat{X}_p}^2) \longrightarrow H^2(\widehat{X}_p; \Omega_{\widehat{X}_p}^2) \right]. \quad (4.5)$$

For a node and for the cones over S_7 and S_6 , contraction with the crepant volume form and the boundary map give natural isomorphisms

$$T_{X,p}^1 \cong K_p^{\text{cr}} \cong H_2(\Sigma_p; \mathbb{C}). \quad (4.6)$$

These isomorphisms commute with the maps to $H^2(\widehat{X}; \Omega_{\widehat{X}}^2)$ and $H_2(\widehat{X}; \mathbb{C})$ induced by the local inclusions. They carry the tangent cone of a smoothing component to $V_{p, \sigma_p} \otimes \mathbb{C}$.

For completeness, the branch assertion is finite. The circle-bundle Gysin sequence identifies the del Pezzo link with $K_{S_p}^\perp$. On the pentagon, the reflection fixing the unique smoothing parameter acts, after the determinant twist from contraction with the volume form, on the unique A_1 root line. On the hexagon, a sixfold rotation has a (-1) eigenspace and a Coxeter eigenspace; these are the A_1 line and A_2 plane. The local cohomology sequence identifies these eigenspaces with (4.5). This proves the branch statement and the compatibility with the Gysin maps; the full derivation is Lemmas 8.6 and 8.7 of [23], which rest on Schlessinger’s comparison [24] in the depth-three form of [25].

4.2 A class of toric anticanonical hypersurfaces

Let $N \cong \mathbb{Z}^4$, let $M = \text{Hom}(N, \mathbb{Z})$, and write $\langle \cdot, \cdot \rangle$ for the natural pairing. For a four-dimensional reflexive polytope $\Delta \subset N_{\mathbb{R}}$, set

$$\Delta^\circ = \{u \in M_{\mathbb{R}} : \langle u, v \rangle \geq -1 \text{ for every } v \in \Delta\}. \quad (4.7)$$

Let Σ_Δ be the fan of cones over the faces of Δ and $\mathbb{P}_\Delta = \text{TV}(\Sigma_\Delta)$ its Gorenstein Fano toric fourfold. A hypersurface $X \subset \mathbb{P}_\Delta$ is Δ -regular if its Laurent equation and every face restriction are nondegenerate in the sense of Batyrev [26].

If $G \subset \Delta$ is a two-face, its dual face $G^* \subset \Delta^\circ$ is an edge. A Δ -regular anticanonical hypersurface meets the torus orbit associated with G in $\ell(G^*)$ points, where ℓ is lattice length; the transverse germ at each point is the affine cone over the toric surface defined by G . We call G *singular* when it is not a unimodular triangle. The unit square, the unit-edge pentagon with one interior lattice point, and the analogous hexagon give, respectively, a node and the cones over S_7 and S_6 .

The explicit finite theorem is proved for the following controlled class. We call Δ admissible if:

1. every two-face is a unimodular triangle, a unit square, or a pentagon or hexagon with one interior lattice point;
2. every edge has lattice length one; and
3. the dual edge of every singular two-face has lattice length one.

For a Δ -regular anticanonical hypersurface, the singular locus is then finite and consists exactly of nodes and cones over S_7 or S_6 . The ambient divisor matrix detects the full rational topological kernel in (4.4); equivalently, the correction term in Batyrev's formula for $h^{1,1}$ [26] vanishes on this class (Proposition 8.3 of [22]).

The abstract local-to-global statement below uses the following additional hypotheses:

$$\omega_X \cong \mathcal{O}_X, \quad h^1(X, \mathcal{O}_X) = 0, \quad \widehat{X} \text{ satisfies the } \partial\bar{\partial}\text{-lemma.} \quad (4.8)$$

The third holds automatically for the projective crepant resolution fixed in Section 3.1; all three hold in the toric hypersurface examples of Section 5.

4.3 The first-order image theorem

Let $\mathbb{T}_X^1$ denote the global first-order deformation space of X . We view $V_{p, \sigma_p} \otimes \mathbb{C}$ as a subspace of $T_{X, p}^1$ through (4.6).

Theorem 4.2 (first-order local-to-global image). *Let X arise from an admissible polytope and satisfy (4.8). For every branch profile σ ,*

$$\text{Im} \left[\mathbb{T}_X^1 \longrightarrow \bigoplus_p T_{X, p}^1 \right] \cap \bigoplus_p (V_{p, \sigma_p} \otimes \mathbb{C}) = K_\sigma(X) \otimes \mathbb{C}. \quad (4.9)$$

Equivalently, in the divisor and root bases of Section 3, the right-hand side is $\ker Q_{X, \sigma}$.

Proof. For a normal Gorenstein threefold with finite non-local-complete-intersection (non-lci) locus, the cotangent complex and Kähler-differential deformation groups agree through degree two (Lemma 8.5 of [23]). The local-to-global Ext sequence therefore begins

$$0 \longrightarrow H^1(X; \Theta_X) \longrightarrow \mathbb{T}_X^1 \xrightarrow{\text{loc}} \bigoplus_p T_{X,p}^1 \xrightarrow{\text{ob}} H^2(X; \Theta_X). \quad (4.10)$$

We identify the kernel of ob on each branch.

Let $E = \text{Exc}(\widehat{X} \rightarrow X)$ and $X_{\text{reg}} = \widehat{X} \setminus E$. Excision and the local-cohomology sequence give the exact segment

$$H^1(X_{\text{reg}}; \Omega_{X_{\text{reg}}}^2) \longrightarrow \bigoplus_p H_{E_p}^2(\widehat{X}_p; \Omega_{\widehat{X}_p}^2) \longrightarrow H^2(\widehat{X}; \Omega_{\widehat{X}}^2). \quad (4.11)$$

By Lemma 4.1, each branch space $V_{p,\sigma_p} \otimes \mathbb{C}$ is a subspace of K_p^{cr} in the middle term. Moreover, the middle map in (4.11) fits into the commutative square

$$\begin{array}{ccc} \bigoplus_p H_2(\Sigma_p; \mathbb{C}) & \longrightarrow & H_2(\widehat{X}; \mathbb{C}) \\ \downarrow \wr & & \downarrow \text{PD} \\ \bigoplus_p K_p^{\text{cr}} & \longrightarrow & H^4(\widehat{X}; \mathbb{C}), \end{array} \quad (4.12)$$

where the lower arrow is the last map of (4.11) followed by the de Rham map. Thus a link class in the topological kernel (4.4) has zero de Rham image. The $\partial\bar{\partial}$ -lemma makes $H^2(\widehat{X}; \Omega_{\widehat{X}}^2) \rightarrow H^4(\widehat{X}; \mathbb{C})$ injective, so exactness of (4.11) lifts that class to $H^1(X_{\text{reg}}; \Omega_{X_{\text{reg}}}^2)$.

Contraction with a trivialization of ω_X identifies $H^1(X_{\text{reg}}; \Omega_{X_{\text{reg}}}^2)$ with $H^1(X_{\text{reg}}; \Theta_{X_{\text{reg}}})$. Schlessinger's depth-three comparison [24], in the form of [25], identifies the latter with $\mathbb{T}_X^1$, and naturality identifies the resulting localization map with loc in (4.10). This proves that every element of $K_\sigma(X) \otimes \mathbb{C}$ lies in the left-hand side of (4.9). Conversely, any global deformation class maps to zero under ob ; the same commutative square sends its link class to zero in $H_2(\widehat{X}; \mathbb{C})$. Lemma 4.1 then identifies its branch restriction with an element of $K_\sigma(X)$. The two inclusions prove the equality. This argument is the kernel comparison of Theorem D (Theorem 8.8) and Section 8.4 of [23], printed here to make the maps and obstruction group explicit. $\square$

Combining Theorem 4.2 with Proposition 3.1 yields the central identification

$$\text{Im}\left(\mathcal{M}_{H,\sigma}^{\text{rigid}} \rightarrow \mathcal{B}_\sigma\right) = \text{Im}\left(\mathbb{T}_X^1 \rightarrow \bigoplus_p T_{X,p}^1\right) \cap \mathcal{B}_\sigma. \quad (4.13)$$

This is the sense in which the geometric kernel is physically derived as a five-dimensional charge and gauging constraint.

4.4 All-orders necessity

A first-order statement does not by itself constrain an analytic arc whose local parameter begins at higher order. The necessary condition for a genuine smoothing is obtained instead from vanishing periods. For a node, a distinguished vanishing three-sphere has period proportional to the local deformation parameter. For the divisorial contractions used here, the Milnor fiber contains a distinguished three-cycle whose period is nonzero on the relevant component. Mayer–Vietoris boundary maps relate these cycles to classes in (4.4). The resulting argument is insensitive to the order of contact with the local discriminant. The required local period statement and its simultaneous full-support consequence are Theorem 6.1 and Remark 6.2, respectively, of [22].

Theorem 4.3 (all-orders necessary relation). *Let X be a compact complex threefold with $\omega_X \cong \mathcal{O}_X$ and $h^1(X, \mathcal{O}_X) = 0$, whose singularities are nodes and anticanonical cones over S_7 or S_6 . Fix a crepant resolution $\hat{X} \rightarrow X$, small over the nodes and with the corresponding del Pezzo surface over each cone point. Let $\mathcal{X} \rightarrow \mathbb{D}$ be a flat proper analytic family over a disc, with central fiber $\mathcal{X}_0 \cong X$ and smooth fibers $\mathcal{X}_s$ for all sufficiently small $s \neq 0$, and let σ be the branch profile it induces: at each E_3 point, the component of the miniversal base through which the family smooths the germ. Then the projection*

$$K_\sigma(X) \longrightarrow V_{q, \sigma_q} \tag{4.14}$$

is nonzero for every node q , every E_2 point q , and every E_3 point q at which $\sigma_q = \mathfrak{sl}_2$.

Remark 4.4. Since $\mathbb{Q}$ is infinite, a generic rational linear combination avoids the finite union of the coordinate hyperplanes in question, so $K_\sigma(X)$ contains a single class whose component is nonzero simultaneously at every germ named in Theorem 4.3. The statement applies to the actual analytic arc and not merely to its first derivative.

Theorem 4.3 gives no conclusion at an E_3 point smoothed through $\mathcal{H}_{\mathfrak{sl}_3}$, although it still constrains every other germ of the same family, with $K_\sigma(X)$ computed on the profile that family induces. That component has a two-dimensional Milnor kernel. The analogous proof requires two independent period pairings, not only a nonzero scalar period, and remains open. The first-order Theorem 4.2 does include this plane.

4.5 Integrability boundary

Theorem 4.2 implies that every branch-restricted class in the kernel lifts to a global first-order deformation. It does not imply that this deformation integrates. The del Pezzo cone germs are not local complete intersections [27], and the standard unobstructedness argument for nodal Calabi–Yau threefolds does not apply uniformly on the non- $\mathbb{Q}$ -factorial locus relevant here.

Three distinct assertions should therefore remain separate:

1. $b \in \ker Q_{X, \sigma}$ solves the projected rigid moment-map equations;

2. b is the localization of a global first-order complex deformation;
3. b is tangent to an actual finite extremal transition.

The first two are proved for every profile in the admissible class. The third is proved for purely nodal transitions and for the explicit X_9 family in Section 5.2; it is open in general. This is the remaining step in a necessary-and-sufficient mixed analogue of Friedman’s conifold-transition criterion [28].

5 Compact examples: forbidden, mixed and obstructed deformations

Three compact configurations exhibit distinct consequences of the charge map: a local Higgs flow that disappears, a mixed transition that exists only through cooperation, and a large system with many partial directions but no simultaneous all-sector trajectory. These effects depend on the actual kernel, not only its dimension.

5.1 A single E_2 direction that disappears globally

The smallest example, denoted X_7 , is a compact anticanonical hypersurface with one isolated cone over S_7 and no other singularities. Locally, the E_2 theory has the reduced Higgs coordinate β of Section 2.3. In the convention (4.7), it is a Δ_7 -regular hypersurface in $\mathbb{P}_{\Delta_7}$, where

$$\begin{aligned} \Delta_7 = \text{conv}\{v_0 = (1, 0, 0, 0), v_1 = (0, 1, 0, 0), v_2 = (0, 0, 1, 0), v_3 = (-6, -4, -1, 0), \\ v_4 = (0, 0, 0, 1), v_5 = (-6, -4, 0, -1), v_6 = (-3, -2, 1, -1)\}. \end{aligned} \quad (5.1)$$

This reflexive polytope has fifteen unimodular triangular two-faces and one unit-edge pentagon $G_7 = \text{conv}\{v_2, v_3, v_4, v_5, v_6\}$ with the unique interior lattice point $p_7 = (-3, -2, 0, 0)$; its dual edge has length one. The three remaining boundary lattice points of Δ_7 lie in facet interiors, so their divisors miss a generic hypersurface. We fix a maximal projective crepant partial (MPCP) subdivision whose restriction to G_7 is the five-triangle star at p_7 . All other two-faces are already unimodular, so this is the only local resolution choice entering the root calculation. Writing D_i for the ray divisor of v_i , the reduced A_1 root pairs with the ambient divisor classes as

$$b_{7,\beta} = -D_2 - D_3 + D_4 + D_5. \quad (5.2)$$

The row is nonzero and the reduced local branch is one-dimensional. Hence the root class has nonzero image in $H_2(\widehat{X}_7; \mathbb{Q})$, and therefore

$$K_{\text{red}}(X_7) = 0. \quad (5.3)$$

The corresponding compact vector gauges the $SU(2)$ Cartan with nonzero coefficient. Its complex moment-map equation sets $\beta = 0$.

Equation (5.3) has two consequences at different logical levels. Theorem 4.2 says that no global first-order complex deformation moves the reduced local E_2 coordinate. The all-orders

Theorem 4.3 then excludes a one-parameter smoothing whose first nonzero local term appears at higher order. Thus a genuine local Higgs branch of the isolated fixed point is absent from the compact deformation space.

This example is intentionally elementary. A single local sector supplies nothing with which its nontrivial root class can cancel. It provides the clean negative control for the cooperative example below.

5.2 A mixed smoothing requiring simultaneous local deformations

The main positive example X_9 contains two nodes and one cone over S_7 . Order the reduced invariant coordinates as

$$x = (u_1, u_2, \beta)^T, \quad (5.4)$$

where u_1, u_2 are conifold moment maps and β is the E_2 Cartan moment map. No nonzero relation among the two nodal curve classes alone has a nonzero coefficient on either curve. The E_2 root likewise cannot move by itself, as the preceding example illustrates.

The underlying reflexive polytope is

$$\begin{aligned} \Delta_9 = \text{conv}\{v_0 = (1, 0, 0, 0), v_1 = (0, 1, 0, 0), v_2 = (-1, -1, 0, 0), \\ v_3 = (0, 0, 1, 0), v_4 = (0, 0, 0, 1), v_5 = (-4, -2, -1, 0), \\ v_6 = (-4, -2, 0, -1), v_7 = (3, 1, 1, 1), v_8 = (2, 1, 1, 1)\}. \end{aligned} \quad (5.5)$$

Its singular two-faces are the squares $\{v_2, v_3, v_6, v_7\}$ and $\{v_2, v_4, v_5, v_7\}$ and the pentagon $\{v_3, v_4, v_5, v_6, v_8\}$. Star subdivision at the pentagon's interior point $p = (-2, -1, 0, 0)$ gives $E = D_p \cong S_7$. The only other boundary lattice point, $(-1, 0, 0, 0)$, lies in a facet interior, and its divisor misses a generic anticanonical hypersurface. We fix an MPCP completion using the square diagonals (v_6, v_7) and (v_2, v_4) , in the order just listed. With the sign convention that the chosen diagonal has coefficient -1 , the two nodal circuit rows on the ten rays $(v_0, \dots, v_8, p)$ are

$$v_2 + v_3 - v_6 - v_7, \quad -v_2 - v_4 + v_5 + v_7. \quad (5.6)$$

The remaining choices in the MPCP completion lie over smooth strata and do not alter these circuit rows or the star of p . Principal-divisor relations eliminate D_0, D_1, D_3, D_4 , giving the divisor basis below.

After removing a toric divisor that does not meet the generic hypersurface, a saturated basis of the toric divisor quotient is

$$\mathcal{D} = (D_2, D_5, D_6, D_7, D_8, E), \quad (5.7)$$

where $E \cong S_7$ is the exceptional divisor. This is a complete basis of $H^2(\widehat{X}_9; \mathbb{Q})$. In this basis,

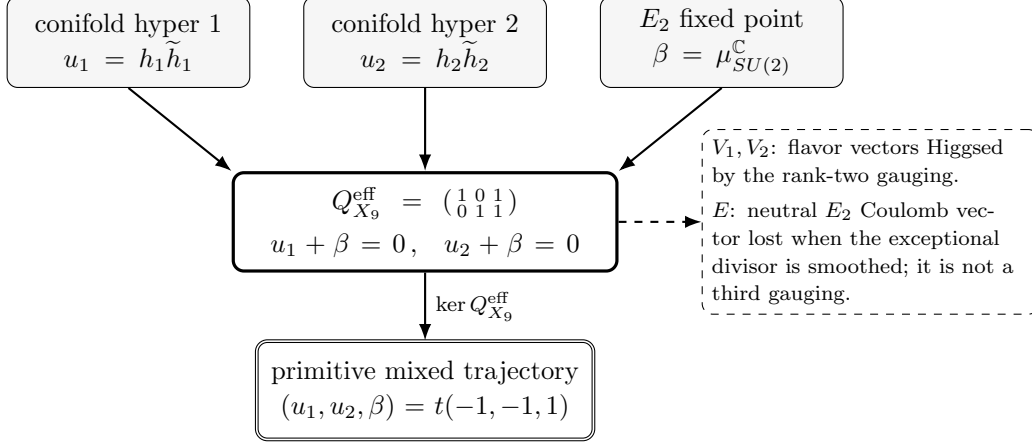


Figure 2. Cooperative Higgsing in X_9 . No nonzero trajectory is supported on the two conifold sectors or on the E_2 sector alone. The unique global direction activates all three, while the protected vector-multiplet count separates the two Higgsed flavor vectors from the neutral SCFT Coulomb vector.

the compact gauging matrix is

$$Q_{\mathcal{D}} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (5.8)$$

An integral unimodular change of the six compact vector directions isolates two effective vectors,

$$V_1 = -D_6, \quad V_2 = D_5, \quad (5.9)$$

and brings (5.8) to

$$Q_{X_9}^{\text{eff}} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}. \quad (5.10)$$

Both nonzero Smith invariants are one in the printed toric lattice. The two complex moment-map equations are

$$u_1 + \beta = 0, \quad u_2 + \beta = 0, \quad (5.11)$$

so

$$\ker Q_{X_9}^{\text{eff}} = \mathbb{C}(-1, -1, 1). \quad (5.12)$$

The cooperative structure is displayed in Figure 2.

No nonzero vector in (5.12) is supported only on the nodes or only on the E_2 sector. The two free-hypermultiplet sectors and the interacting sector can move only together. Conversely, an explicit one-parameter Calabi–Yau family realizes the class

$$(u_1, u_2, \beta) = t(-1, -1, 1) \quad (5.13)$$

and has smooth general fiber by Theorem B of [23]. More explicitly, take the facet $F = \{v_2, v_3, v_4, v_5, v_6, v_7, v_8\}$ and deformation degree $R = (0, 1, -1, -1)$. On the ordered edges

$$(v_2v_5, v_2v_6, v_2v_7, v_3v_6, v_3v_7, v_3v_8, v_4v_5, v_4v_7, v_4v_8, v_5v_6, v_7v_8) \quad (5.14)$$

the edge-dilation vector is $\tau = (0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0)$. This is the compatible Minkowski decomposition used in the Ilten–Vollmert construction [29, 30]: it restricts to a segment-plus-segment decomposition on each square and to a segment-plus-unimodular-triangle decomposition on the pentagon. These data define the cited toric deformation and its relative anticanonical family. The kernel is therefore not only a first-order flat direction: it integrates to a mixed extremal transition, and it is cooperative in the sense defined in the Introduction.

There is a useful spectrum check. Exact lattice-point and Mayer–Vietoris calculations give

	$h^{1,1}$	$h^{2,1}$	χ	n_V	n_H^{neutral}
$\widehat{X}_9$	6	122	-232	5	123
$X_{9,t}$	3	123	-240	2	124

(5.15)

For a smooth compact Calabi–Yau threefold in M-theory, $n_V = h^{1,1} - 1$ and $n_H^{\text{neutral}} = h^{2,1} + 1$. The two effective gaugings in (5.10) Higgs two vector directions and remove two quaternionic directions from the product of local Higgs sectors. One quaternionic modulus remains, in agreement with the increase of n_H^{neutral} in (5.15).

The transition loses three, not two, massless vector multiplets. The third has a different origin. The exceptional-divisor vector E is neutral under the E_2 root moment map because

$$E|_E = K_E, \quad \beta \in K_E^\perp. \quad (5.16)$$

It is the rank-one E_2 Coulomb direction and disappears when the entire surface is replaced by the smoothing. It is not a third flavor gauging. This separation between Higgsed vectors and the lost SCFT Coulomb vector is needed for the protected spectrum to agree with the topology.

The integrality statement above is made in the saturated toric divisor lattice. A possible finite index between that lattice and the full integral cohomology lattice would affect global forms or discrete remnants, but not the rank, the complex kernel or the multiplet count. We also do not claim the full massive spectrum at the interacting E_2 point.

5.3 Thirty locally smoothable sectors with no common trajectory

The large example X° contains

$$26 \text{ nodes}, \quad 2 \text{ } E_2 \text{ points}, \quad 2 \text{ } E_3 \text{ points}. \quad (5.17)$$

Every singular germ admits a local smoothing. At the i th E_2 point, let α_i denote the nilpotent scheme-theoretic tangent coordinate and β_i the reduced smoothing coordinate. At the j th E_3 point, let L_j be the A_1 coordinate and (H_{j1}, H_{j2}) the A_2 coordinates. Record a profile by

profile	$\dim_{\mathbb{C}} \mathcal{B}_{\sigma}$	$\text{rank } Q_{\sigma}$	$\dim_{\mathbb{C}} \ker Q_{\sigma}$	forced reduced coordinates
(2, 2)	30	21	9	$u_9, u_{11}, \beta_1, \beta_2$
(2, 3)	31	21	10	$u_9, u_{11}, \beta_1, \beta_2, L_1$
(3, 2)	31	21	10	$u_9, u_{11}, \beta_1, \beta_2, L_2$
(3, 3)	32	20	12	$u_9, u_{11}, \beta_1, \beta_2$

Table 3. The four branch-restricted compact charge maps for X° . In a profile (σ_1, σ_2) , $\sigma_j = 2$ selects the $\mathfrak{sl}_2$ line and $\sigma_j = 3$ the $\mathfrak{sl}_3$ plane at the j th E_3 point; α_i is the nilpotent scheme-theoretic tangent coordinate and β_i the reduced smoothing coordinate at the i th E_2 point, while L_j is the $\mathfrak{sl}_2$ line coordinate at the j th E_3 point. Every nonzero Smith invariant is one in the printed divisor lattice.

the ordered pair of component labels selected at the two E_3 points. The four reduced profiles are

$$(2, 2), \quad (2, 3), \quad (3, 2), \quad (3, 3), \quad (5.18)$$

where 2 and 3 are subscripts naming the selected simple factor: 2 selects the $\mathfrak{sl}_2$ line and 3 the $\mathfrak{sl}_3$ plane.

Appendix B.2 gives the defining fan polytope Π , all singular two-faces, a strict convex height function fixing the crepant resolution, and the complete sparse charge matrix. Thus X° denotes a specified Π -regular anticanonical hypersurface rather than only an abstract singularity inventory.

In an integral divisor frame the full local-to-global matrix has size 36×26 and rank 21; the physical charge matrix is its transpose. Restricting to the four profiles gives the exact data in Table 3.

The forced-zero conclusions are witnessed by individual compact vectors, not only by a rank calculation. Five primitive integral vector combinations have complex moment maps

$$\begin{aligned}
\mu_{V_9}^{\mathbb{C}} &= u_9 - \alpha_1 + \alpha_2, & \mu_{V_{11}}^{\mathbb{C}} &= u_{11} - \alpha_1 + \alpha_2, \\
\mu_{V_{\beta_1}}^{\mathbb{C}} &= \beta_1, & \mu_{V_{\beta_2}}^{\mathbb{C}} &= \alpha_2 + \beta_2, \\
\mu_{V_{A_1}}^{\mathbb{C}} &= L_1 + L_2.
\end{aligned} \quad (5.19)$$

At an ordinary complex point of the E_2 Higgs scheme, $\alpha_1 = \alpha_2 = 0$. The first four equations in (5.19) therefore force

$$u_9 = u_{11} = \beta_1 = \beta_2 = 0 \quad (5.20)$$

on every reduced profile. The fifth equation correlates L_1 and L_2 on (2, 2), isolates the remaining line coordinate on (2, 3) or (3, 2), and becomes inactive on (3, 3). This explains the rank drop in the last row of Table 3.

The correct conclusion is not that X° has no Higgs directions. Its projected kernels have dimensions between nine and twelve. Rather, no profile contains a point at which all thirty local sectors have nonzero smoothing coordinates. The all-orders result is stronger than this first-order observation. Any one-parameter smoothing of X° would smooth the

nodes N_9 and N_{11} and both E_2 points, whatever components it selects at the two E_3 points. Theorem 4.3 and Remark 4.4 then require a class in the kernel of the induced profile with nonzero projection to the coordinates in (5.20), and Table 3 excludes such a class on every profile. Consequently no analytic one-parameter family smooths all thirty germs, even if its local parameters begin at different orders. This is Theorem 6.5 of [22], whose Corollary 6.7 also excludes every formal arc meeting all thirty local smoothing loci.

5.4 Finite classification for polytopes with at most nine vertices

The three examples are part of a finite, reproducible classification. Among the 77 admissible reflexive four-polytopes with at most nine vertices in the Kreuzer–Skarke list [31] and a degree-seven cone point, the all-orders necessity condition obstructs 76; 67 of these have a single singular point, while nine have one node in addition to the E_2 point. The remaining case is X_9 , whose explicit mixed family is smooth. Thus cooperative Higgsing is the unique smoothable outcome in this precisely stated bounded class by Proposition 9.2 and Corollary 6.7 of [22] and Theorems B and C of [23].

This census shows that the compact constraint is systematic rather than an accident of one matrix. It is not a statistical sample of all compact Calabi–Yau threefolds, and no universality claim is inferred from the ratio 76/77.

6 Normalizability and gravity-decoupling limits

The compact theorem and a rigid noncompact limit impose different normalizability conditions. In the compact model, the rows of Q_X arise from dynamical vector fields. In a noncompact limit, some of those fields can become background flavor sources, in which case their moment-map equations must be removed. We formulate the distinction in this section.

6.1 Finite compact volume

On a smooth compact resolution, the standard reduction (3.2) gives the five-dimensional kinetic term

$$S_5 \supset -\frac{1}{4} \int_{M_5} G_{IJ} F^I \wedge *_5 F^J, \quad G_{IJ} \propto \int_{\widehat{X}} \omega_I \wedge *_6 \omega_J. \quad (6.1)$$

[32]. Every harmonic form on a compact manifold has finite norm. After separating the graviphoton, these fields supply the $h^{1,1}(\widehat{X}) - 1$ vector multiplets. At the transition point the graviphoton direction is the Kähler class of the singular threefold, which pairs to zero with every exceptional curve and root, so its row of Q_X vanishes and the effective rows in Section 5 are vector-multiplet directions. The pairings in (3.5) and (3.9) are therefore charges under dynamical fields, not charges under nondynamical flavor backgrounds.

Similarly, the kinetic metric for compact complex-structure moduli is schematically

$$G_{\kappa\bar{\lambda}} \propto \frac{\int_X \chi_\kappa \wedge \bar{\chi}_{\bar{\lambda}}}{\int_X \Omega \wedge \bar{\Omega}}, \quad \chi_\kappa \in H^{2,1}(X). \quad (6.2)$$

[32]. The local coordinates in $\mathcal{B}_\sigma$ are limits of these compact directions, subject to the global image theorem. Thus no additional local L^2 assumption is needed to impose

$$Q_X b = 0 \tag{6.3}$$

at finite compact volume.

6.2 Dynamical and background vectors in a local limit

Consider a one-parameter family of Kähler classes $J(\Lambda)$ with

$$\text{Vol}_{J(\Lambda)}(X) \longrightarrow \infty, \quad G_{N,5} \longrightarrow 0. \tag{6.4}$$

For a vector combination $v = v^I D_I$, let $\omega_v = v^I \omega_I$ and define its kinetic norm

$$\mathcal{N}_v(\Lambda) = \int_X \omega_v \wedge *_{J(\Lambda)} \omega_v, \tag{6.5}$$

understood in a finite-normalization basis after the standard five-dimensional Einstein-frame rescaling. The vector belongs to the retained dynamical theory only when its normalized kinetic term has a finite nonzero limit. Geometrically, this is the requirement that it become an L^2 two-form on the retained noncompact space. A compact divisor may furnish such a form; a divisor which becomes noncompact gives a non-normalizable flavor background.

Let P_{dyn} project the compact vector lattice onto the combinations that remain dynamical, and let $\mathcal{B}_{\text{loc}}$ be the product of the local invariant-coordinate spaces retained in the chosen noncompact limit. The moment-map matrix of the limiting theory is

$$\boxed{Q_{\text{dyn}} = P_{\text{dyn}} Q_X, \quad \text{Im}(\mathcal{M}_H \rightarrow \mathcal{B}_{\text{loc}}) = \ker Q_{\text{dyn}}.} \tag{6.6}$$

If the norm of a shared vector diverges, its canonically normalized coupling to the local sector vanishes. That vector is no longer integrated over in the local theory; its moment-map equation is not imposed, and the corresponding row of Q_X is deleted. Equation (6.6) is therefore the limit of the compact derivation, not an independent ansatz.

6.3 Multistage decoupling

Several local singularities can remain connected after the overall Calabi–Yau volume is sent to infinity. Shared compact surfaces or lower-codimension singular loci can continue to gauge a common flavor subgroup. Only a further decompactification factorizes the individual local theories. This multistage structure appears in geometric constructions of 5d theories and in compact analyses of generalized symmetries [16, 17].

For the present mechanism there are accordingly three regimes:

regime	dynamical shared vectors	constraint
compact supergravity	all compact rows	$Q_X b = 0$
glued local QFT	selected finite-norm rows	$P_{\text{dyn}} Q_X b = 0$
factorized local sectors	no shared rows	no cross-sector equation.

(6.7)

Coupling to gravity is therefore sufficient but not necessary for the correlation. A gravity-decoupled glued QFT may retain it as an ordinary 5d gauging. What is special to the compact statement is that every row is dynamical before a limit is chosen.

6.4 Complex deformations and the local L^2 caveat

The same distinction applies to Higgs coordinates. Local geometric-engineering work uses a protected notion of dynamical complex deformation that is checked against Higgs chiral rings and magnetic quivers [12, 15]. Strict L^2 normalizability of a representative three-form can additionally depend on the asymptotic metric of the chosen noncompact Calabi–Yau, with subtleties already visible for the conifold. We therefore separate:

1. compact normalizability, which is sufficient for the theorem here;
2. protected local Higgs dynamics, established for the node, E_2 and E_3 components used above; and
3. strict local L^2 realization in a specified noncompact metric, which is not proved in this paper.

No Kähler scaling ray has yet been constructed for X_9 or X° which determines exactly which shared rows survive after gravity decouples. Accordingly, our principal theorem concerns compact M-theory at finite volume. Claims about a particular rigid limit must use (6.6) and supply that additional geometric analysis.

7 Discussion

The compact selection rule has a simple physical origin. Local Higgs sectors carry electric charges or flavor moment maps. A compact Calabi–Yau does not leave all of the corresponding flavor symmetries global: its C_3 vectors gauge diagonal combinations determined by divisor–curve pairings. Their moment-map equations cut out a correlated subspace of the product of local deformation coordinates. The same divisor–curve map appears in the local-to-global deformation sequence, so the five-dimensional vacuum equation and the geometric smoothing relation have the same kernel.

For nodes this reduces to the familiar conifold transition. The substantive extension is that an interacting del Pezzo sector enters the relation on the same footing as a wrapped-M2 hypermultiplet, through a current-multiplet moment map rather than an elementary field bilinear. The X_9 relation

$$(u_1, u_2, \beta) = t(-1, -1, 1) \tag{7.1}$$

is the simplest realization: two conifold condensates and an E_2 Higgs operator participate in one primitive compact transition. Neither part exists as a global transition without the other.

The E_3 sector shows why a branchwise formulation is necessary. Its local Higgs branch is not irreducible. The components $\mathcal{H}_{\mathfrak{sl}_3}$ and $\mathcal{H}_{\mathfrak{sl}_2}$ couple to different compact cocharacters, have different dimensions, and can change the rank of the global gauging matrix. Replacing the local theory by $\dim T^1 = 3$ would erase precisely the field-theoretic information responsible for the profile dependence of X° .

7.1 Relation to generalized-symmetry gluing

Compact M-theory models also correlate the generalized symmetries and global forms of several local sectors. There, Mayer–Vietoris maps determine which defects and higher-form symmetries glue consistently [17]. The analogy with the present result is structural. In both cases, independent local data are replaced by the kernel or cokernel of a map supplied by the complement of the singular neighborhoods in the compact threefold.

The constrained objects are different. Generalized-symmetry calculations track defect groups, polarizations and higher-group structures. Our matrix acts on ordinary Higgs moment maps and complex deformation coordinates. One result does not follow from the other. Together they indicate that the global topology of a compactification controls more than the spectrum of isolated local theories: it also controls how their moduli and symmetry data can be assembled.

7.2 Comparison with the compact del Pezzo–conifold precursor

Malyshev established that a root curve in a compact del Pezzo surface can be homologous to curves at spatially separated conifold points, and that this relation reduces the number of del Pezzo deformations [13]. That calculation proves the geometric phenomenon in a particular quintic. It does not derive the relation as a five-dimensional gauging or formulate a theorem for interacting Higgs branches.

The branchwise charge-kernel theorem is therefore a strict extension, both in scope and in physical content. It applies uniformly to the admissible toric class and identifies the relation matrix with electric and flavor gaugings. It retains the reduced and embedded E_2 schemes and both E_3 components, determines the exact image of the moment-map locus, and separates first-order lifting from all-orders necessity. The positive and negative examples include an independent protected-spectrum check. The antecedent is the mixed homology relation; the new result is its branchwise five-dimensional kernel theorem.

7.3 Open problems

Several extensions would sharpen the result.

1. *Integrability.* The principal mathematical problem is to decide whether every full-support class in $K_\sigma(X)$ integrates on the non- $\mathbb{Q}$ -factorial locus. A positive answer would give a mixed necessary-and-sufficient criterion generalizing the nodal theorem.

2. *The E_3 plane at all orders.* The first-order theorem includes $\mathcal{H}_{\mathfrak{sl}_3}$. The period proof should be extended from one distinguished period to a rank-two pairing detecting both A_2 directions.
3. *Controlled rigid limits.* Explicit Kähler rays for X_9 and X° would determine whether any of their shared gaugings survives after gravity decouples. This would turn the general projection $P_{\text{dyn}}Q_X$ into a concrete glued-QFT construction.
4. *Integral global form.* The examples determine primitive Smith data in their toric divisor lattices. Establishing index one in full integral cohomology would settle the corresponding discrete gauge remnants and line operators.
5. *The compact scalar geometry.* Equation (4.13) identifies a protected holomorphic projection. Constructing the full coupled quaternionic-Kähler space and its chiral-ring analogue would go beyond the rigid zero-level description.
6. *Broader sectors.* Non-isolated singularities, higher-rank 5d SCFTs and non-toric compact Calabi–Yau threefolds should admit analogous maps, but their branch lattices and local Hodge comparisons require new input.

The mixed-transition mechanism already follows from the results above. Higgs directions that are independent in isolated local theories need not remain independent in a compact M-theory vacuum. Dynamical five-dimensional vectors select their allowed combinations, and a global transition can require cooperation between free conifold states and an interacting SCFT sector. This is the compact generalization of the conifold charge relation.

A Local invariant rings and balanced representatives

This appendix records two local calculations used in Proposition 3.1.

A.1 E_3 maximal-torus invariants

The closure of the minimal nilpotent orbit of $\mathfrak{sl}_n$ has the rank-one presentation

$$\overline{\mathcal{O}_{\min}(\mathfrak{sl}_n)} = \text{Spec} \left(\frac{\mathbb{C}[q_1, \dots, q_n, \tilde{q}_1, \dots, \tilde{q}_n]}{(\sum_i q_i \tilde{q}_i)} \right)^{\mathbb{C}_{\text{ADHM}}^*}, \quad (\text{A.1})$$

where the ADHM torus acts with weights $+1$ and -1 on q_i and $\tilde{q}_i$. The complex moment-map matrix is

$$\Phi^i_j = q_i \tilde{q}_j, \quad \text{tr } \Phi = 0. \quad (\text{A.2})$$

The flavor torus acts by $q_i \mapsto t_i q_i$ and $\tilde{q}_i \mapsto t_i^{-1} \tilde{q}_i$, with $\prod_i t_i = 1$. After combining it with the ADHM torus, an invariant monomial has equal powers of q_i and $\tilde{q}_i$ for each i . The invariant ring is therefore generated by

$$z_i = q_i \tilde{q}_i, \quad \sum_i z_i = 0. \quad (\text{A.3})$$

Since a complex torus is linearly reductive, taking invariants is exact and introduces no additional relation. Thus

$$\mathbb{C}[\overline{\mathcal{O}_{\min}(\mathfrak{sl}_3)}]^{T_{SU(3),\mathbb{C}}} \cong \frac{\mathbb{C}[z_1, z_2, z_3]}{(z_1 + z_2 + z_3)} \cong \mathbb{C}[H_1, H_2], \quad (\text{A.4})$$

$$\mathbb{C}[\overline{\mathcal{O}_{\min}(\mathfrak{sl}_2)}]^{T_{SU(2),\mathbb{C}}} \cong \frac{\mathbb{C}[w_1, w_2]}{(w_1 + w_2)} \cong \mathbb{C}[L]. \quad (\text{A.5})$$

One convenient root-basis convention is

$$H_1 = z_1, \quad H_2 = -z_3, \quad L = w_1 = -w_2. \quad (\text{A.6})$$

The E_3 chiral ring is the fiber product of the two orbit rings over their origins. Its invariant ring is consequently

$$\mathbb{C}[H_1, H_2] \times_{\mathbb{C}} \mathbb{C}[L] \cong \frac{\mathbb{C}[H_1, H_2, L]}{(H_1 L, H_2 L)}. \quad (\text{A.7})$$

A.2 Balanced real moment maps

For arbitrary $(H_1, H_2) \in \mathbb{C}^2$, solve (A.3) and (A.6) by

$$z_1 = H_1, \quad z_2 = -H_1 + H_2, \quad z_3 = -H_2. \quad (\text{A.8})$$

For each $z_i = |z_i|e^{i\theta_i}$ choose

$$q_i = \sqrt{|z_i|}e^{i\theta_i}, \quad \tilde{q}_i = \sqrt{|z_i|}. \quad (\text{A.9})$$

Then $q_i \tilde{q}_i = z_i$ and $|q_i|^2 - |\tilde{q}_i|^2 = 0$ for every i . The ADHM real moment map and all flavor-Cartan real moment maps vanish. The $SU(2)$ component is identical with $w_1 = L, w_2 = -L$. Equations (A.9), (2.3) and (2.12) supply the local balanced lifts used in the proof of Proposition 3.1.

A.3 Abelian monopole sums for the E_3 components

Removing the diagonal magnetic charge from the affine- A_1 quiver leaves $m \in \mathbb{Z}$. Its monopole sum is

$$\frac{1}{1-x} \sum_{m \in \mathbb{Z}} x^{|m|} = \frac{1+x}{(1-x)^2}. \quad (\text{A.10})$$

For the affine- A_2 triangle, set one diagonal charge to zero. The sum is

$$\frac{1}{(1-x)^2} \sum_{m_1, m_2 \in \mathbb{Z}} x^{\frac{1}{2}(|m_1 - m_2| + |m_1| + |m_2|)} = \frac{1+4x+x^2}{(1-x)^4}. \quad (\text{A.11})$$

The two series are respectively the character dimensions of the $SU(2)$ representations of highest weight $2n$ and the $SU(3)$ representations with Dynkin labels (n, n) . Adding the two component rings and subtracting their common constant gives (2.23).

B Integral charge data for the compact examples

B.1 The X_9 divisor reduction

In the ordered basis (5.7), the three local coordinate rows are

$$B_{\text{red}} = \begin{pmatrix} 1 & 0 & -1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \end{pmatrix}, \quad Q_{\mathcal{D}} = B_{\text{red}}^{\text{T}}. \quad (\text{B.1})$$

The following integral vector combinations define a unimodular basis:

$$\begin{aligned} V_1 &= -D_6, & V_2 &= D_5, \\ S_1 &= D_2 + D_7, & S_2 &= D_2 + D_5 + D_6, \\ S_3 &= D_8, & \Phi &= E. \end{aligned} \quad (\text{B.2})$$

The determinant of the change-of-basis matrix is -1 , and in this frame

$$Q' = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (\text{B.3})$$

This proves both the effective matrix and its primitivity in the printed toric lattice.

The full E_2 tangent scheme also contains the nilpotent coordinate α . Its row is the pairing of the divisors with a class of $K_{S_7}^\perp$ complementary to the root; the choice of complement is immaterial on the reduced branch, where $\alpha = 0$. In the divisor order (5.7), the row is

$$(0, -2, 1, 0, -2, 0). \quad (\text{B.4})$$

Some spectator combinations can therefore couple to α over dual numbers. This does not add a finite Higgs mass because $\alpha^2 = \alpha\beta = 0$ and α vanishes on the reduction.

B.2 The X° toric and resolution input

Let $N_\circ \cong \mathbb{Z}^4$ be a fresh one-parameter-subgroup lattice and set $\Pi = \text{conv}\{r_1, \dots, r_{26}\} \subset (N_\circ)_\mathbb{R}$. Thus Π is the fan polytope: the ambient fourfold is defined by the fan over its faces, and X° is a Π -regular anticanonical hypersurface. In particular, Π is not the polar $\Delta^\circ \subset M_\mathbb{R}$

of (4.7). The following table is the complete vertex list; its order fixes all subsequent labels.

1	(-1, -1, -1, -1)	14	(0, -1, -1, 0)
2	(-1, -1, -1, 0)	15	(0, -1, 0, 1)
3	(-1, -1, 0, -1)	16	(0, -1, 0, 0)
4	(-1, -1, 0, 1)	17	(0, 1, -1, -1)
5	(-1, -1, 1, 0)	18	(0, 1, -1, 0)
6	(-1, -1, 1, 1)	19	(0, 1, 0, -1)
7	(-1, 0, -1, -1)	20	(0, 1, 0, 0)
8	(-1, 0, -1, 0)	21	(1, 1, -1, -1)
9	(-1, 0, 0, -1)	22	(1, 0, -1, 0)
10	(-1, 0, 0, 1)	23	(0, 0, 0, -1)
11	(-1, 0, 1, 0)	24	(0, 0, 0, 1)
12	(-1, 0, 1, 1)	25	(0, 0, 1, 0)
13	(0, -1, -1, -1)	26	(0, 0, 1, 1)

(B.5)

The twenty-six unit-square faces, written by vertex index, are

N_1	{1, 2, 7, 8}	N_{14}	{5, 6, 25, 26}
N_2	{1, 2, 13, 14}	N_{15}	{7, 8, 17, 18}
N_3	{1, 3, 7, 9}	N_{16}	{7, 9, 17, 19}
N_4	{2, 4, 8, 10}	N_{17}	{10, 12, 18, 20}
N_5	{2, 4, 14, 15}	N_{18}	{10, 12, 24, 26}
N_6	{3, 5, 9, 11}	N_{19}	{11, 12, 19, 20}
N_7	{3, 5, 13, 16}	N_{20}	{11, 12, 25, 26}
N_8	{3, 5, 23, 25}	N_{21}	{13, 14, 15, 16}
N_9	{3, 9, 19, 23}	N_{22}	{13, 16, 23, 25}
N_{10}	{4, 6, 10, 12}	N_{23}	{15, 16, 25, 26}
N_{11}	{4, 10, 15, 24}	N_{24}	{17, 18, 19, 20}
N_{12}	{5, 6, 11, 12}	N_{25}	{18, 20, 24, 26}
N_{13}	{5, 6, 15, 16}	N_{26}	{19, 20, 25, 26}

(B.6)

Each set in (B.6) denotes the corresponding vertices r_i . The remaining singular faces are

$D_{6,1}$	{ $r_1, r_2, r_3, r_4, r_5, r_6$ }	$r_{27} = (-1, -1, 0, 0)$
$D_{6,2}$	{ $r_7, r_8, r_9, r_{10}, r_{11}, r_{12}$ }	$r_{28} = (-1, 0, 0, 0)$
$D_{7,1}$	{ $r_1, r_7, r_{13}, r_{17}, r_{21}$ }	$r_{29} = (0, 0, -1, -1)$
$D_{7,2}$	{ $r_2, r_8, r_{14}, r_{18}, r_{22}$ }	$r_{30} = (0, 0, -1, 0),$

(B.7)

where the last column is the unique interior lattice point. These face data give precisely the singularity inventory in (5.17).

For definiteness, one smooth projective crepant subdivision is fixed as follows. On the thirty boundary rays assign heights

$$b_i = 2^{26-i} \quad (1 \leq i \leq 26), \quad b_i = -2^{30} + 2^{30-i} \quad (27 \leq i \leq 30). \quad (\text{B.8})$$

Take the lower-facet triangulation on every facet and cone it to the origin. The induced triangulations agree on common faces, every maximal cone is unimodular, and the four divisorial faces are star-triangulated at $r_{27}, \dots, r_{30}$. This fixes the exceptional curves and surfaces used below.

Principal-divisor relations eliminate the rays 15, 22, 24, 26. Thus the rank-26 divisor basis is indexed by

$$\mathcal{I} = \{1, 2, \dots, 14, 16, 17, 18, 19, 20, 21, 23, 25, 27, 28, 29, 30\}. \quad (\text{B.9})$$

For a local coordinate a , write its charge row as $b_a = \sum_{I \in \mathcal{I}} B_{aI} D_I$. The complete sparse presentation of B is

$$\begin{aligned} b_{N_1} &= D_1 - D_2 - D_7 + D_8, & b_{N_2} &= D_1 - D_2 - D_{13} + D_{14}, \\ b_{N_3} &= D_1 - D_3 - D_7 + D_9, & b_{N_4} &= D_2 - D_4 - D_8 + D_{10}, \\ b_{N_5} &= D_2 - D_4 - D_{14}, & b_{N_6} &= D_3 - D_5 - D_9 + D_{11}, \\ b_{N_7} &= D_3 - D_5 - D_{13} + D_{16}, & b_{N_8} &= D_3 - D_5 - D_{23} + D_{25}, \\ b_{N_9} &= D_3 - D_9 + D_{19} - D_{23}, & b_{N_{10}} &= D_4 - D_6 - D_{10} + D_{12}, \\ b_{N_{11}} &= D_4 - D_{10}, & b_{N_{12}} &= D_5 - D_6 - D_{11} + D_{12}, \\ b_{N_{13}} &= D_5 - D_6 - D_{16}, & b_{N_{14}} &= D_5 - D_6 - D_{25}, \\ b_{N_{15}} &= D_7 - D_8 - D_{17} + D_{18}, & b_{N_{16}} &= D_7 - D_9 - D_{17} + D_{19}, \\ b_{N_{17}} &= D_{10} - D_{12} - D_{18} + D_{20}, & b_{N_{18}} &= D_{10} - D_{12}, \\ b_{N_{19}} &= D_{11} - D_{12} - D_{19} + D_{20}, & b_{N_{20}} &= D_{11} - D_{12} - D_{25}, \\ b_{N_{21}} &= D_{13} - D_{14} - D_{16}, & b_{N_{22}} &= D_{13} - D_{16} - D_{23} + D_{25}, \\ b_{N_{23}} &= -D_{16} + D_{25}, & b_{N_{24}} &= D_{17} - D_{18} - D_{19} + D_{20}, \\ b_{N_{25}} &= D_{18} - D_{20}, & b_{N_{26}} &= D_{19} - D_{20} - D_{25}, \\ b_{D_{6,1},s_1} &= D_1 - D_2 - D_5 + D_6, & b_{D_{6,1},s_2} &= D_2 - D_3 - D_4 + D_5, \\ b_{D_{6,1},s_3} &= D_1 - D_2 - D_3 + D_4 + D_5 - D_6, & b_{D_{6,2},s_1} &= D_7 - D_8 - D_{11} + D_{12}, \\ b_{D_{6,2},s_2} &= D_8 - D_9 - D_{10} + D_{11}, & b_{D_{6,2},s_3} &= D_7 - D_8 - D_9 + D_{10} + D_{11} - D_{12}, \\ b_{D_{7,1},\alpha} &= -2D_1 + 2D_7 + D_{13} - D_{17}, & b_{D_{7,1},\beta} &= -D_1 + D_{13} + D_{17} - D_{21}, \\ b_{D_{7,2},\alpha} &= 2D_2 - D_8 - 2D_{14}, & b_{D_{7,2},\beta} &= D_8 - D_{14} - D_{18}. \end{aligned}$$

This is a 36×26 matrix of rank 21, with twenty-one nonzero Smith invariants, all equal to one. Deleting the coordinates absent on a chosen E_3 component gives the four profile matrices in Table 3.

B.3 Primitive vectors in X°

If B denotes the 36×26 local-coordinate-by-divisor matrix, the five primitive vectors producing (5.19) are

$$\begin{aligned}
v_9 &= \sum_{i=1}^{12} D_i + D_{17} + D_{18} + D_{19} + D_{20}, \\
v_{11} &= - \sum_{i=7}^{12} D_i - D_{17} - D_{18} - D_{19} - D_{20} - D_{21}, \\
v_{\beta_1} &= -D_{21}, \\
v_{\beta_2} &= \sum_{i=1}^{12} D_i - D_{21}, \\
v_{A_1} &= D_1 + D_3 + D_5 + D_7 + D_9 + D_{11} + D_{13} + D_{16} + D_{17} + D_{19} + D_{21} + D_{23} + D_{25}.
\end{aligned} \tag{B.10}$$

Direct multiplication gives

$$\begin{aligned}
Bv_9 &= e_{u_9} - e_{\alpha_1} + e_{\alpha_2}, & Bv_{11} &= e_{u_{11}} - e_{\alpha_1} + e_{\alpha_2}, \\
Bv_{\beta_1} &= e_{\beta_1}, & Bv_{\beta_2} &= e_{\alpha_2} + e_{\beta_2}, \\
Bv_{A_1} &= e_{L_1} + e_{L_2}.
\end{aligned} \tag{B.11}$$

The complete sparse matrix is included in the reproducibility source described in Appendix C. Its Smith form has 21 nonzero unit invariants and five zero directions. After branch restriction, the four profiles have the ranks printed in Table 3.

C Reproducibility

All computations used in the physics translation are exact integer, rational or symbolic calculations. No numerical fit or floating-point rank decision enters a claim. From the project root, the local checks are run with

```

venv/bin/python checks/check_nodal_charge.py
venv/bin/python checks/check_e2_gauging.py
venv/bin/python checks/check_e3_gauging.py
venv/bin/python checks/check_e3_magnetic_quivers.py
venv/bin/python checks/check_xcirc_e3_profiles.py
venv/bin/python checks/check_x9_spectrum.py

```

The scripts verify, respectively:

1. the nodal moment-map image and charge-kernel identity;
2. the E_2 invariant ring, balanced lift and X_9 reduced matrix;

3. the E_3 invariant ring, root basis and branchwise balanced lifts;
4. the affine- A_1/A_2 monopole sums and fiber-product Hilbert-series identity;
5. the four X° profile ranks, Smith invariants and isolating vectors; and
6. the X_9 divisor quotient, Hodge numbers, effective vectors and multiplet changes.

The source geometric computations are maintained with the companion mathematical papers. For the thirty-sector obstruction they include the independent reconstructions of the 36×26 matrix, its topological kernel, the local Milnor kernels and the vanishing-period statements. For X_9 they include the singular-locus calculation, branch restriction, explicit family, divisor quotient and Batyrev lattice-point counts [22, 23].

The finite scripts check the printed matrices and formulas. They do not replace the Hodge-theoretic proof of Theorem 4.2, the period proof of Theorem 4.3, or the geometric construction of the X_9 family.

The manuscript source and exact finite checks are available at <https://github.com/bwuebben/global-higgs-branch>.

References

- [1] B.R. Greene, D.R. Morrison and A. Strominger, *Black hole condensation and the unification of string vacua*, *Nucl. Phys. B* **451** (1995) 109 [[hep-th/9504145](#)].
- [2] S. Katz, D.R. Morrison and M.R. Plesser, *Enhanced gauge symmetry in type II string theory*, *Nucl. Phys. B* **477** (1996) 105 [[hep-th/9601108](#)].
- [3] E. Witten, *Phase transitions in M-theory and F-theory*, *Nucl. Phys. B* **471** (1996) 195 [[hep-th/9603150](#)].
- [4] T. Mohaupt and F. Saueressig, *Effective supergravity actions for conifold transitions*, *JHEP* **03** (2005) 018 [[hep-th/0410272](#)].
- [5] N. Seiberg, *Five dimensional SUSY field theories, non-trivial fixed points and string dynamics*, *Phys. Lett. B* **388** (1996) 753 [[hep-th/9608111](#)].
- [6] D.R. Morrison and N. Seiberg, *Extremal transitions and five-dimensional supersymmetric field theories*, *Nucl. Phys. B* **483** (1997) 229 [[hep-th/9609070](#)].
- [7] M.R. Douglas, S. Katz and C. Vafa, *Small instantons, del Pezzo surfaces and type I' theory*, *Nucl. Phys. B* **497** (1997) 155 [[hep-th/9609071](#)].
- [8] K. Intriligator, D.R. Morrison and N. Seiberg, *Five-dimensional supersymmetric gauge theories and degenerations of Calabi–Yau spaces*, *Nucl. Phys. B* **497** (1997) 56 [[hep-th/9702198](#)].
- [9] S. Cremonesi, G. Ferlito, A. Hanany and N. Mekareeya, *Instanton operators and the Higgs branch at infinite coupling*, *JHEP* **04** (2017) 042 [[1505.06302](#)].
- [10] G. Ferlito and A. Hanany, *A tale of two cones: The Higgs branch of $Sp(n)$ theories with $2n$ flavours*, [1609.06724](#).

- [11] S. Cabrera, A. Hanany and F. Yagi, *Tropical geometry and five dimensional Higgs branches at infinite coupling*, *JHEP* **01** (2019) 068 [[1810.01379](#)].
- [12] C. Closset, S. Schäfer-Nameki and Y.-N. Wang, *Coulomb and Higgs branches from canonical singularities: Part 0*, *JHEP* **02** (2021) 003 [[2007.15600](#)].
- [13] D. Malyshev, *del Pezzo singularities and SUSY breaking*, *Adv. High Energy Phys.* **2011** (2011) 630892 [[0705.3281](#)].
- [14] A. Bourget, J.F. Grimminger, A. Hanany, M. Sperling and Z. Zhong, *Magnetic quivers from brane webs with $O5$ planes*, *JHEP* **07** (2020) 204 [[2004.04082](#)].
- [15] M. De Marco, M. Del Zotto, J. Grimminger and A. Sangiovanni, *5d Higgs branches: Stratifications from geometry*, [2607.06851](#).
- [16] F. Apruzzi, S. Schäfer-Nameki and Y.-N. Wang, *5d SCFTs from decoupling and gluing*, *JHEP* **08** (2020) 153 [[1912.04264](#)].
- [17] M. Cvetič, J.J. Heckman, M. Hübner and E. Torres, *Generalized symmetries, gravity, and the swampland*, *Phys. Rev. D* **109** (2024) 026012 [[2307.13027](#)].
- [18] A. Collinucci, M. De Marco, A. Sangiovanni and R. Valandro, *Higgs branches of 5d rank-zero theories from geometry*, *JHEP* **10** (2021) 018 [[2105.12177](#)].
- [19] K. Altmann, *The versal deformation of an isolated toric Gorenstein singularity*, *Invent. Math.* **128** (1997) 443.
- [20] S. Cremonesi, A. Hanany and A. Zaffaroni, *Monopole operators and Hilbert series of Coulomb branches of $3d \mathcal{N} = 4$ gauge theories*, *JHEP* **01** (2014) 005 [[1309.2657](#)].
- [21] A. Ceresole and G. Dall’Agata, *General matter coupled $N = 2$, $D = 5$ gauged supergravity*, *Nucl. Phys. B* **585** (2000) 143 [[hep-th/0004111](#)].
- [22] B.J. Wuebben, “A vanishing-cycle obstruction to smoothing Calabi–Yau threefolds.” Preprint, available at <https://github.com/bwuebben/calabi-yau-smoothability>, 2026.
- [23] B.J. Wuebben, “Smoothing Calabi–Yau threefolds with nodes and del Pezzo cone points.” Preprint, available at <https://github.com/bwuebben/calabi-yau-smoothability>, 2026.
- [24] M. Schlessinger, *Rigidity of quotient singularities*, *Invent. Math.* **14** (1971) 17.
- [25] R. Friedman and R. Laza, *Deformations of singular Fano and Calabi–Yau varieties*, *J. Differential Geom.* **131** (2025) 65 [[2203.04823](#)].
- [26] V.V. Batyrev, *Dual polyhedra and mirror symmetry for Calabi–Yau hypersurfaces in toric varieties*, *J. Algebraic Geom.* **3** (1994) 493 [[alg-geom/9310003](#)].
- [27] M. Gross, *Deforming Calabi–Yau threefolds*, *Math. Ann.* **308** (1997) 187 [[alg-geom/9506022](#)].
- [28] R. Friedman, *Simultaneous resolution of threefold double points*, *Math. Ann.* **274** (1986) 671.
- [29] K. Altmann, *Minkowski sums and homogeneous deformations of toric varieties*, *Tohoku Math. J.* **47** (1995) 151.
- [30] N.O. Ilten and R. Vollmert, *Deformations of rational T -varieties*, *J. Algebraic Geom.* **21** (2012) 531 [[0903.1393](#)].
- [31] M. Kreuzer and H. Skarke, *Complete classification of reflexive polyhedra in four dimensions*, *Adv. Theor. Math. Phys.* **4** (2000) 1209 [[hep-th/0002240](#)].

- [32] A.C. Cadavid, A. Ceresole, R. D'Auria and S. Ferrara, *11-dimensional supergravity compactified on Calabi–Yau threefolds*, *Phys. Lett. B* **357** (1995) 76 [[hep-th/9506144](#)].